

MICRO-DEGREE

Künstliche Intelligenz und Gesellschaft

***Artificial Intelligence
and Society***



Micro-Degree Artificial Intelligence and Society

– Technical Aspects of AI –

Lecture 4.2: Predicting Sequences

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Self-Supervised Learning for Sequence Prediction

- **Self-supervised learning:** a form of **unsupervised learning** in which not humans but the data itself provides the supervision for the learning.
- **Sequence prediction:** given an existing sequence (observation) with its elements (attributes), **predict the next element** of the sequence (output value).
- Two prominent application examples.
 - **Translation:** The input sequence is the text in one language, the output sequence is the same text but in a different language.
 - **Text generation:** The input sequence is a prompt, the output sequence is the continuation of the prompt.

Translation

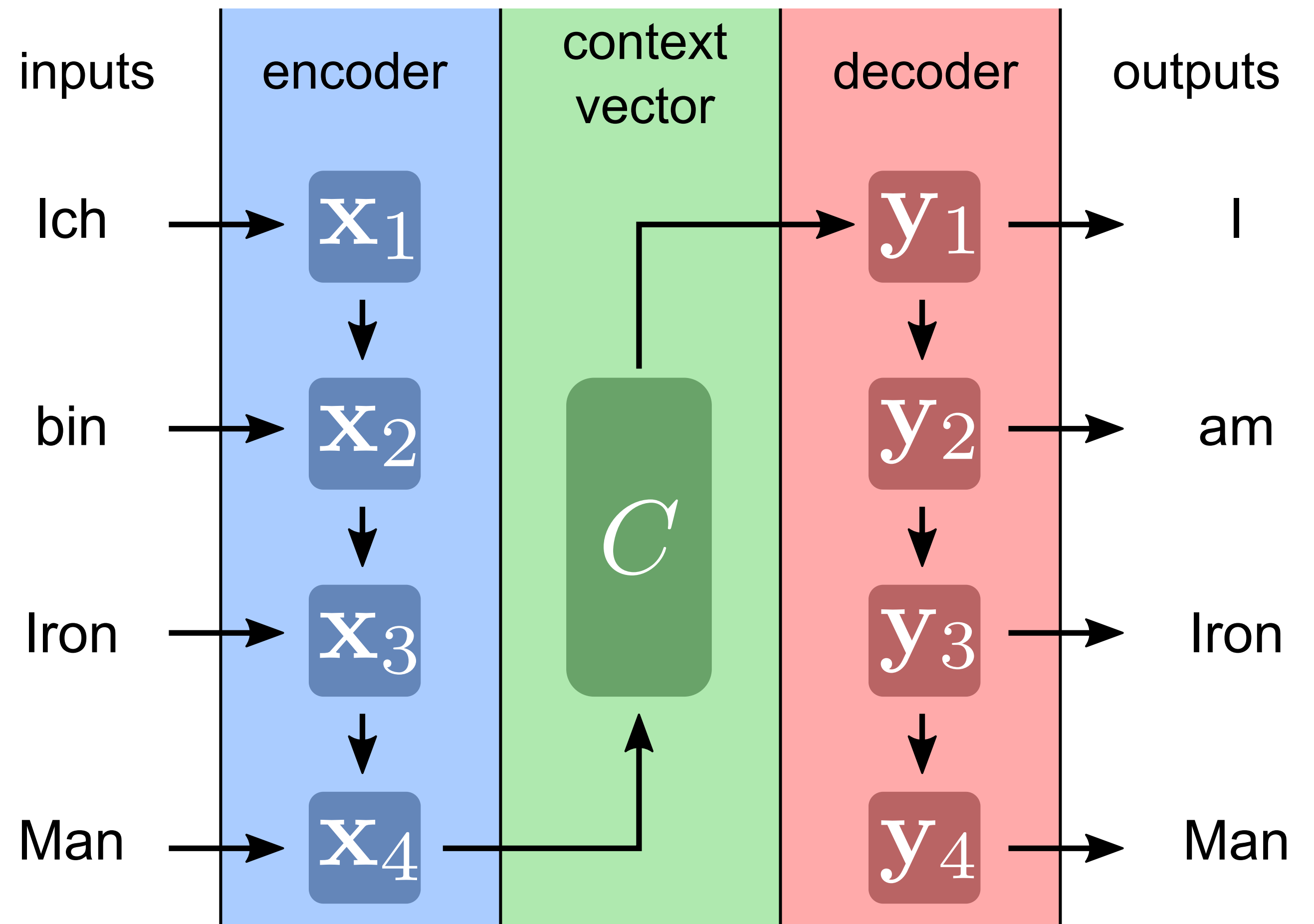
Encoder – Decoder Architecture

Task

Given a sequence of words in one language, predict a sequence of words in another language.

Challenges

- Sequential computation
- Limited context length



Traditional Encoder-Decoder Architecture for Translation.

Source: Adapted from [CS 198-126: Lecture 14 - Transformers and Attention](#)

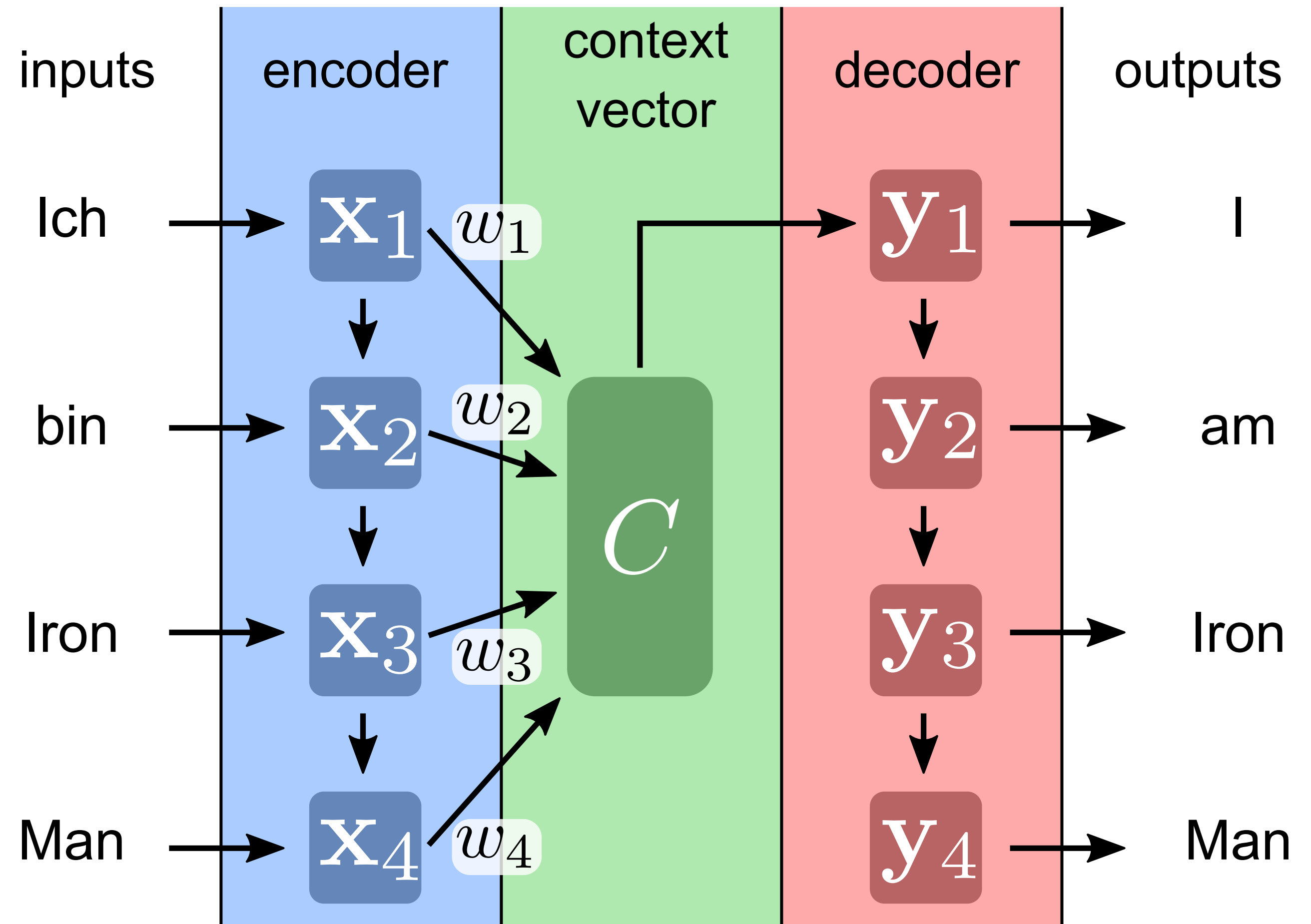
Encoder – Decoder with "Attention"

Attention

Learn the importance of each word in context using weights w_i .

Advantages

- Parallel computation
- Infinite context length



Traditional Encoder-Decoder Architecture for Translation.

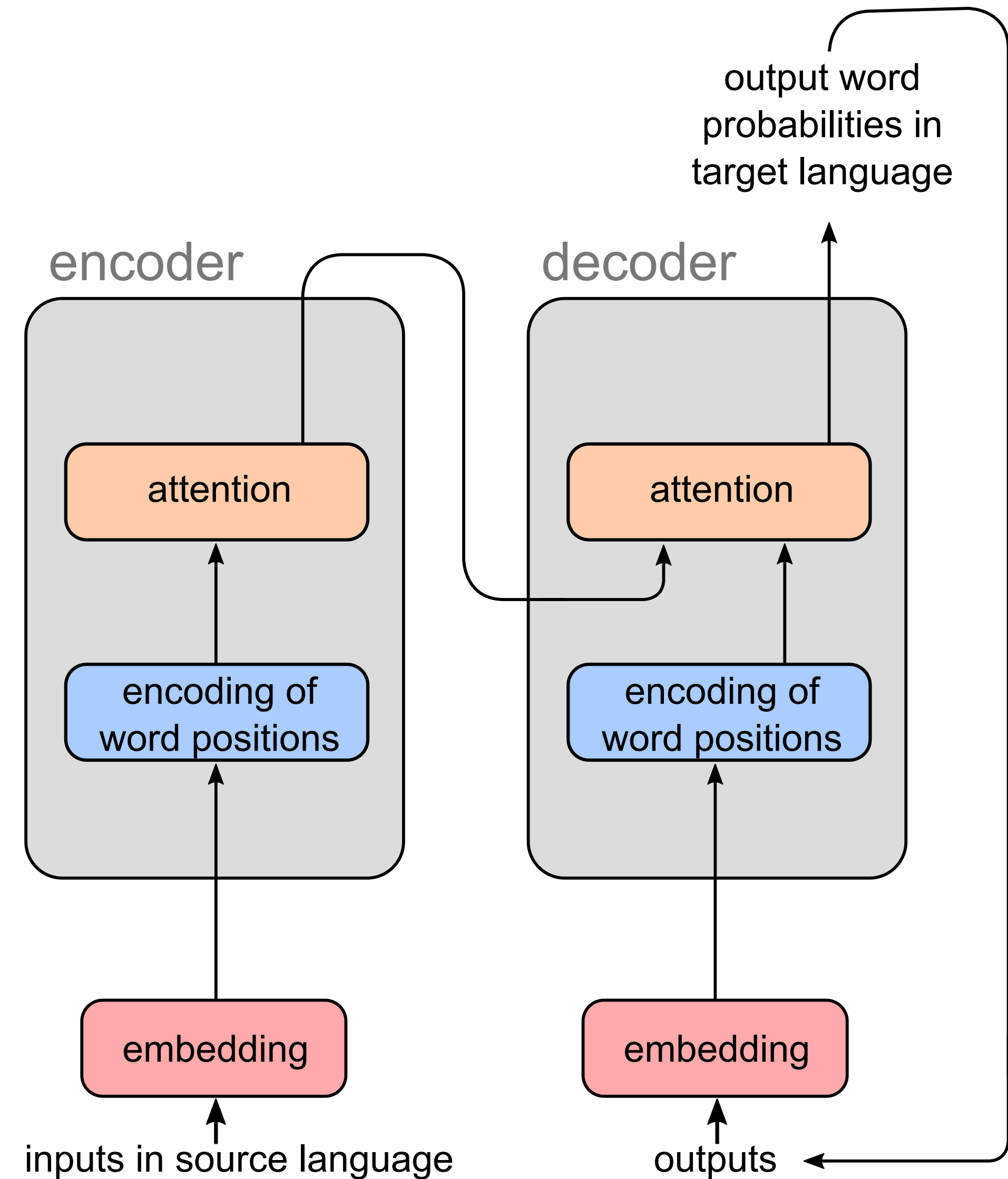
Source: Adapted from [CS 198-126: Lecture 14 - Transformers and Attention](#)

"Transformer" – Architecture for Translation

Figure adapted from Vaswani et al.
[Attention is all you need](#) (2017).

Every predicted word depends on two types of input:

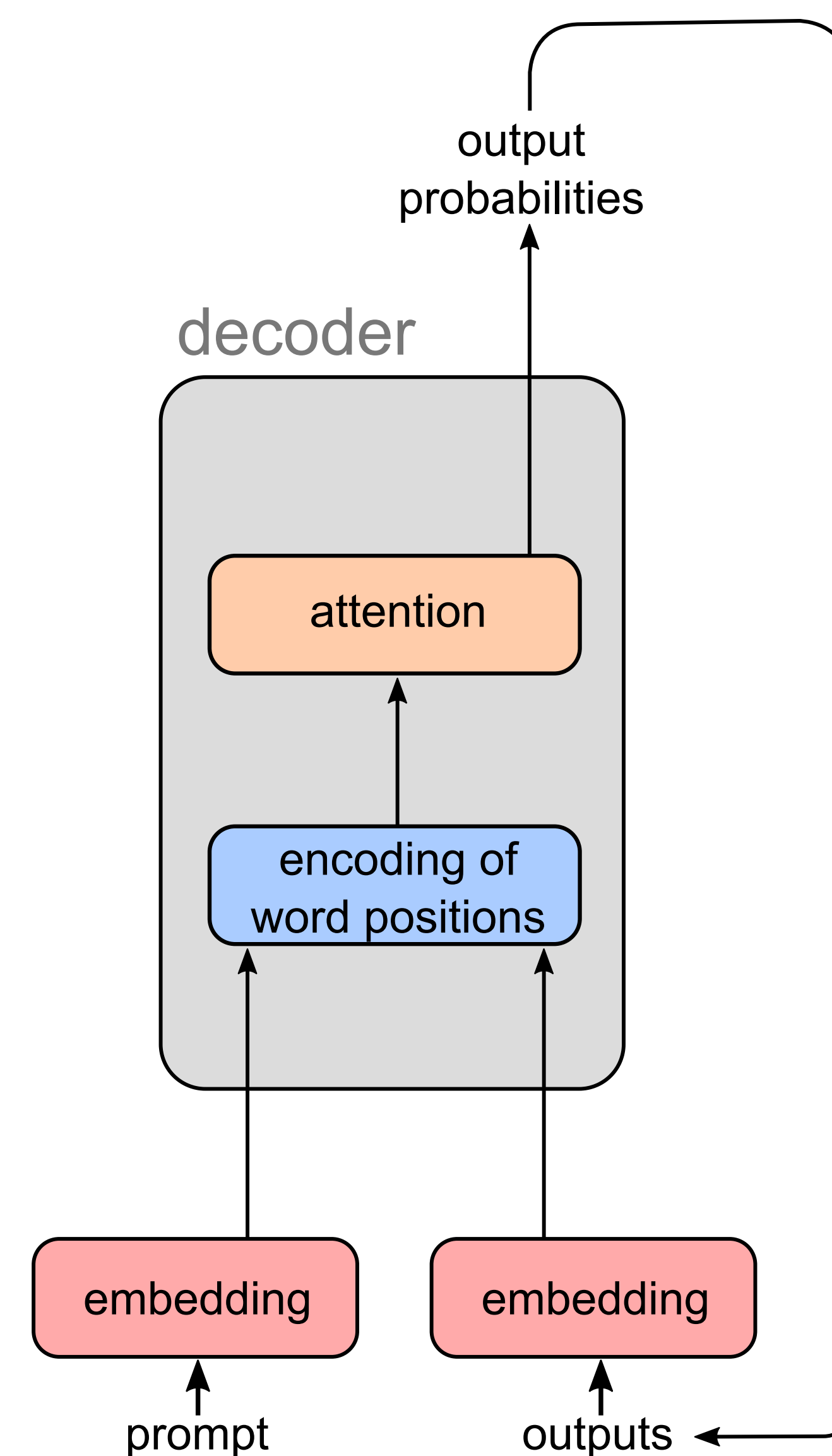
- The full sequence in the source language.
- The already generated (partial) sequence in the target language.



Text generation

"Transformer" – Architecture for Text Generation

Every predicted word depends only on the words preceding it, including the prompt.



Conditional Probabilities

Goal: given a sequence of words, predict the next word

I didn't have a coffee today yet. Because of this I am ...

angry

tired

yellow

$$P(\text{Rain}) = 0.5$$

$$P(\text{rain} \mid \text{weather report}) = 0.8$$

$$P(\text{rain} \mid \text{weather report, clouds}) = 0.9$$

$$P(\text{angry} \mid I, \text{had, no, ...}) = 0.001$$

$$P(\text{tired} \mid I, \text{had, no, ...}) = 0.01 \quad \text{"context"}$$

$$P(\text{yellow} \mid I, \text{had, no, ...}) = 0.0001$$

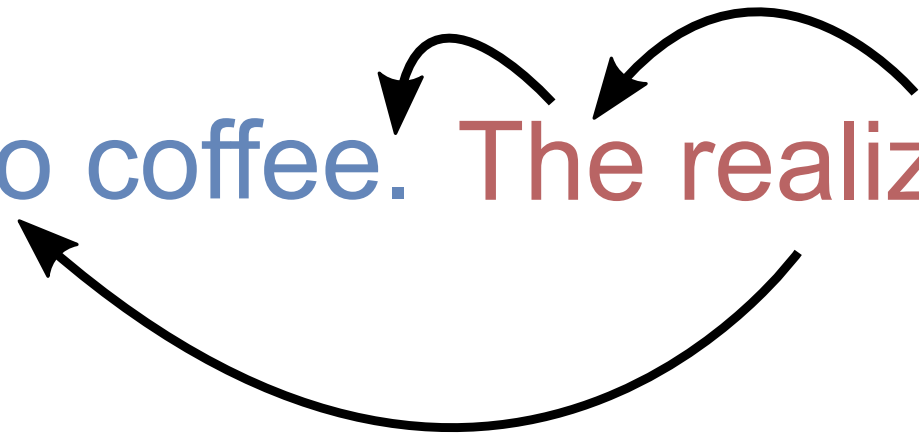
The relationships between sequences and probabilities for the next word are learned from vast amounts of text.

If $P(\text{tired})$ is greater than all other probabilities, then predict "tired".

Intuition for "Attention"

Task: Given a sequence of words, generate the next word.

This morning, I had no coffee. The realization

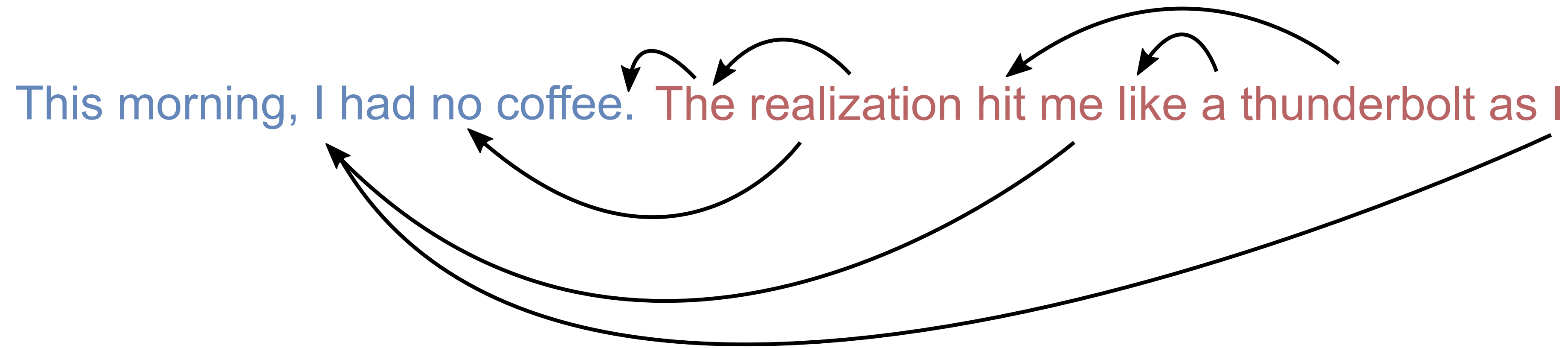


The diagram illustrates the concept of attention in a sequence-to-sequence model. It shows the text "This morning, I had no coffee. The realization". The words "The realization" are highlighted in red. Three curved arrows originate from the words "The", "realization", and "The" and point back to the word "coffee", indicating that the model is paying attention to the context provided by the words preceding the current word being generated.

While the text is being generated, **attention** is paid to different parts of the input.

Intuition for "Attention"

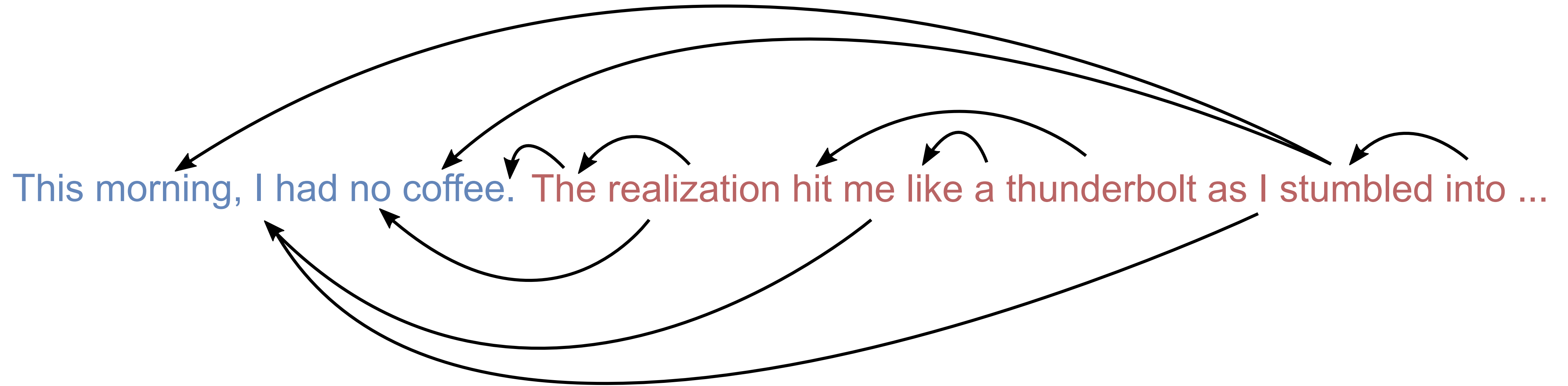
Task: Given a sequence of words, generate the next word.



How much attention each word in the input receives is **learned** during training.

Intuition for "Attention"

Task: Given a sequence of words, generate the next word.



This approach enables interactions across long text segments.

Summary

- To build an algorithm that can predict the next element in a sequence we use **self-supervised learning** (target values are provided by the data itself):
 - The observation is the input sequence. The attributes of the observation are the individual elements of the sequence. The target value is the next element in the sequence.
 - Each predicted element is added to the observation before the next element is predicted.
 - Relationships between preceding elements and the next probable element are learned.
- When using **attention**, it is also learned how important a preceding element is. This enables **longe range interactions** between elements in the input sequence and the target value.
- As primary application examples for sequence prediction we discussed **machine translation** and **text generation**.
- The architecture of neural nets for sequence prediction is called **transformer**. For text translation it consists of an encoder and a decoder, for text generation it is decoder only.